Lecture 39: Transformer- Voltage Regulation and Efficiency

Transformer Voltage Regulation & Efficiency:

Voltage regulation and efficiency of transformer are two important parameters of transformer, to characterize its performance.

Voltage Regulation (VR): Voltage regulation is defined as change in secondary voltage when full load is thrown off with primary voltage held constant. (Full load → No load)

$$\%VR = \frac{E_2 - V_{2\,rated}}{V_{2(rated)}} \quad \text{or} \quad \frac{V_{2NL} - V_{2\,rated}}{V_{2\,rated}} \;;\; V_{2NL} = E_2$$

Remember: $\bar{E}_2 = \bar{V}_2 + \bar{I}_2(r_2 + jX_2)$

This voltage regulation is called voltage regulation up. Voltage regulation down refers change in secondary voltage when load changes from No load to full load. The No load voltage is that rated voltage, and when load is connected the voltage is V_2, then

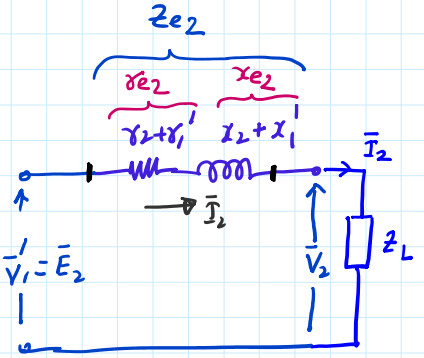
$$\%VR_{down} = \frac{V_{NL} - V_2}{V_{NL}} = \frac{E_2 - V_2}{E_2}$$

Where $V_2 = V_{2\,rated}$ = Secondary voltage at full load

E_2 = Secondary voltage at no load

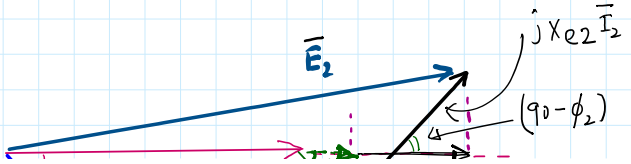
$$Ze_2 = r_{e2} + jX_{e2} = (r_2 + r_1') + j(X_2 + X_1')$$

$$\bar{V}_1' = \bar{E}_2 = \bar{V}_2 + \bar{I}_2 Ze_2 = \bar{V}_2 + \bar{I}_2(r_{e2} + jX_{e2})$$



$$E_2 \approx V_2 + I_2 r_{e2}\cos\phi_2 + I_2 X_{e2}\sin\phi_2$$

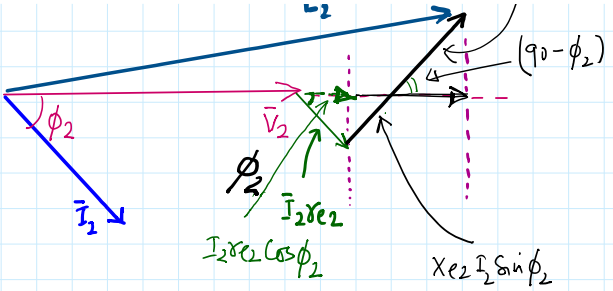
$$E_2 - V_2 = I_2 r_{e2}\cos\phi_2 + I_2 X_{e2}\sin\phi_2$$



$$E_2 - V_2 = I_2 r_{e2} \cos\phi_2 + I_2 x_{e2} \sin\phi_2$$

$$\frac{E_2 - V_2}{V_2} = \frac{I_2 r_{e2} \cos\phi_2 + I_2 x_{e2} \sin\phi_2}{V_2}$$

$$= \left(\frac{I_2 r_{e2}}{V_2}\right)\cos\phi_2 + \left(\frac{I_2 x_{e2}}{V_2}\right)\sin\phi_2$$



$$VR = \frac{E_2 - V_2}{V_2} = r_{e2\,pu}\cos\phi_2 + x_{e2\,pu}\sin\phi_2$$

Ideally VR should be zero. This happens when,

$$r_{e2\,pu}\cos\phi_2 = -x_{e2\,pu}\sin\phi_2$$

$$\text{or } \tan\phi_2 = -\frac{r_{e2}}{x_{e2}}$$

$$\boxed{\phi_2 = -\tan^{-1}\frac{r_{e2}}{x_{e2}}}$$

ϕ_2 negative means, $\bar{I}_2$ leading to V_2 or leading pf. This implies capacitor load. Therefore for a light capacitor load, voltage regulation can be zero, or negative for large capacitor load.

However, load is normally inductive hence VR is positive finite value.

For a practical transformer, the VR should be as small as possible, say 5-10%.

Efficiency of Transformer

The efficiency of a transformer is defined as,

$$\eta = \frac{\text{Output power}}{\text{Input power}} = \frac{P_o}{P_{in}} = \frac{P_2}{P_1}$$

$$= \frac{V_2 I_2 \cos\phi_2}{V_1 I_1 \cos\phi_1} = \frac{V_2 I_2 \cos\phi_2}{\underbrace{V_2 I_2 \cos\phi_2}_{\text{output}} + \underbrace{I_2^2 r_{e2} + P_c}_{\text{Losses}}}$$

$$\text{Input} = \text{output} + \text{Losses}$$

Copper losses = Ohmic losses = Variable osses = $I_2^2 r_{e2}$

Core losses = No load losses = Fixed losses

$$P_c = P_i + P_h$$

Core losses = No load losses = Fixed losses

$P_c = P_h + P_e$

= Hysteresis Current loss + Eddy current loss

Now the question is when this efficiency is maximum as function of load?

Let us differentiate η as function of load current...

$$\frac{d\eta}{dI_2} = \frac{V_2\cos\phi_2\,(V_2 I_2\cos\phi_2 + P_c + I_2^2 r_{e2}) - V_2 I_2\cos\phi_2\,(V_2\cos\phi_2 + 2I_2 r_{e2})}{(V_2 I_2\cos\phi_2 + P_c + I_2^2 r_{e2})^2} = 0$$

$$\Rightarrow \quad V_2\cos\phi_2\,(P_c + I_2^2 r_{e2}) = 2V_2 I_2^2\cos\phi_2 r_{e2}$$

$$\Rightarrow \quad I_2^2 r_{e2} = P_c$$

Variable loss = Fixed loss

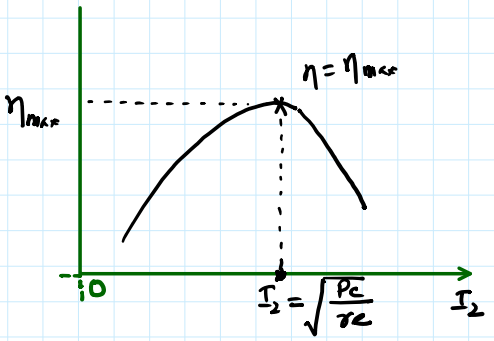
or Copper loss = Core loss

Under this condition: $I_2 = \sqrt{\frac{P_c}{r_{e2}}}$

It can be seen that for this condition, $\frac{d^2\eta}{dI_2^2}$ is -ve thus ensuring $\eta = \eta_{max}$.

The value of η_{max} is given as following:

$$\eta_{max} = \frac{V_2 I_2\cos\phi_2}{V_2 I_2\cos\phi_2 + 2I_2^2 r_{e2}}$$



Example: A 20 KVA, 2500/500 V, 50 Hz, Single-phase transformer has following parameters:

HV Winding

$r_1 = 2.5\ \Omega$

$x_1 = 10\ \Omega$

LV Winding

$r_2 = 0.1\ \Omega$

$x_2 = 0.4\ \Omega$

a) Find the voltage regulation for a power factor of
i) 0.8 pf lagging
ii) 0.8 pf leading
with primary voltage is held at constant at 2500 V.

b) Calculate the efficiency at $\frac{1}{2}$ load, full load at power factor 0.8(lag) and no load losses as 200 W.

c) Calculate the % rated load current at which maximum efficiency takes place. Compute the maximum efficiency.

Solution: (a) Voltage Regulation

$E_2 - V_2 \simeq I_2 (r_{e_2} \cos\phi_2 + x_{e_2} \sin\phi_2)$ for ϕ_2 is +ve for lagging power factor

$I_2 = \frac{(KVA)}{V_2} = \frac{20 \times 1000}{500} = 40\,A = I_2$

$r_{e_2} = r_2 + r_1' = 0.1 + \left(\frac{N_2}{N_1}\right)^2 * r_1$ $\quad \frac{N_2}{N_1} = \frac{500}{2500} = \frac{1}{5}$

$= 0.1 + \frac{1}{25} * 2.5 \; (= 0.1)$

$r_{e_2} = 0.2\,\Omega$

$x_{e_2} = x_2 + x_1' = 0.4 + \left(\frac{1}{25}\right) * 10$

$x_{e_2} = 0.8\,\Omega$

$\phi_2 = \cos^{-1}(0.8) =$

$\cos\phi_2 = 0.8 (lag)$
$\sin\phi_2 = 0.6$

Thus $E_2 - V_2 \simeq 40 (0.2 * 0.8 + 0.8 \times 0.6)$

$E_2 - V_2 = 25.6\,V$

$VR\% = \frac{E_2 - V_2}{V_{rated}} \times 100 = \frac{25.6}{500} \times 100 = 5.12\%$

ii) pf = 0.8 (leading): $E_2 - V_2 \simeq 40 (0.2 \times 0.8 - 0.8 * 0.6)$

$= -12\,V.$

$\%VR = \frac{E_2 - V_2}{V_{2\,rated}} = \frac{-12}{500} = -2.4\%$

(b) $\eta_{half\,load} = \frac{V_2 I_2 \cos\phi_2}{V_2 I_2 \cos\phi_2 + I_2^2 \times r_{e_2} + P_c} = \frac{500 \times 40 \times 0.8}{500 \times 40 \times 0.8 + \left(\frac{40}{2}\right)^2 * 0.2 + 200} \times 100$

(b) $\eta_{halfload} = \dfrac{V_2 I_2 \cos\phi_2 + I_2^2 \times r_{e2} + P_c}{V_2 I_2 \cos\phi_2 + I_2^2 \times r_{e2} + P_c} = \dfrac{500 \times 40 \times 0.8 + \left(\frac{40}{2}\right)^2 * 0.2 + 200}{500 \times 40 \times 0.8 + \left(\frac{40}{2}\right)^2 * 0.2 + 200} \times 100$

$I_{2\,rated} = 40\,A$

$I_{rated\ 1/2} = 20\,A$

$= 96.61\%$

$\eta_{fullload} = \dfrac{500 \times 40 \times 0.8}{500 \times 40 \times 0.8 + (40)^2 * 0.2 + 200} \times 100 = 96.83\%$

(c) η_{max} ?

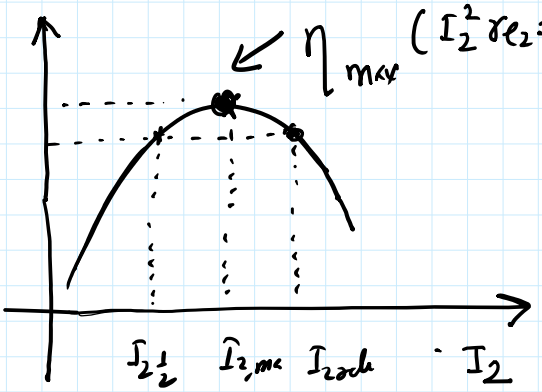
Variable losses = Fixed losses

$I_2^2 r_{e2} = P_c$

$I_2 = \sqrt{\dfrac{P_c}{r_{e2}}} = \sqrt{\dfrac{200}{0.2}} = 31.62\,A$

$\%\,load = \dfrac{31.62}{40} \times 100 = 79\%$ of full load.

$\eta_{max} = \dfrac{500 \times 40 \times 0.8}{500 \times 40 \times 0.8 + 200 + 200} \times 100 = 96.93 \simeq 97\%$



Q10. The maximum efficiency of a 100 kVA, single-phase transformer is 98% and occurs at 80% of full load at 0.8 p.f. If the leakage impedance of the transformer is 5%, find the voltage regulation at rated load of 0.8 p.f. lagging.

Ans.: 3.75%

10. $\eta_{max} = 0.98$, $n = 0.8$

$P_o = 100 \times 1000 \times 0.8 = 80{,}000$ W

$0.98 = \dfrac{80000 \times 0.8}{0.8 \times 80000 + 2P_{core}} \Rightarrow P_{core} = P_{Ohmic} = 653.06$ W

P_{Ohmic} at full load $= \left(\dfrac{100}{80}\right)^2 \times 653.06 = 1020$ W

Transformer voltage information is not given, therefore solving in p.u. terms
By considering $I_2 = 1$ p.u. & $V_2 = 1$ p.u.

$P_{Ohmic\ fl\ (p.u.)} = \dfrac{1020}{100000} = \dfrac{I_2^2 r_{e2}}{V_2 I_2} = \dfrac{r_{e2}}{(V_2/I_2)} = \dfrac{r_{e2}}{Z_{base}}$ Taking $S_{base} = 100$ KVA

$\therefore$ r_{e2} (p.u.) $= 0.0102$ p.u.

x_{e2} (p.u.) $= 0.05$ p.u.

pf $= 0.8$ lag , $\cos\theta_2 = 0.8$, $\sin\theta_2 = 0.6$

$\dfrac{E_2 - V_2}{V_2} = \dfrac{I_2(r_{e2}\cos\theta_2 + x_{e2}\sin\theta_2)}{V_2}$

$VR = r_{e2}\cos\theta_2 + x_{e2}\sin\theta_2$

$= 0.03816$ or 3.816 %